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WEAR

Wear of hot rolling mill rolls: an overview

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Abstract

Rolling is today one of the most important industrial processes because a greater volume of material is worked by rolling than by any other technique. Roll wear is a multiplex process where mechanical and thermal fatigue combines with impact, abrasion, adhesion and corrosion, which all depend on system interactions rather than material characteristics only. The situation is more complicated in section rolling because of the intricacy of roll geometry. Wear variables and modes are reviewed along with published methods and models used in the study and testing of roll wear. This paper reviews key aspects of roll wear control – roll material properties, roll pass design, and system factors such as temperature, loads and sliding velocity. An overview of roll materials is given including adamites, high Cr materials, high speed tool steels and compound rolls. Non-uniform wear, recognised as the most detrimental phenomenon in section rolling, can be controlled by roll pass design. This can be achieved by computer-aided graphical and statistical analyses of various pass series. Preliminary results obtained from pilot tests conducted using a two-disc hot wear rig and a scratch tester are discussed.

Keywords: Rolling; Milling

1. Introduction

Rolling steel at elevated temperatures ("rolling", hereafter) is one of the most important industrial processes, for a greater volume of material is worked by rolling than by any other technique [1,2]. Key tools in the rolling process are the rolls themselves. These tools have to withstand severe extremes of temperature and load. In addition to the obvious need for resistance to breakage, there is the continuing component of roll wear that is critical to the economics of fabrication, and the geometrical tolerances of the rolled products.

From the earliest days of metal working by rolling (some 500 years ago), rolls were used as plain (flat) or as grooved (calibre) rolls [3]. Rolling between flat rolls enables the shaping of rectangular profiles only. Rolling with grooved rolls enables the sophisticated and straightforward manufacture of an enormous variety of profiles. A disadvantage of calibre rolls, supplementary to complications in their production and maintenance, is the significantly higher wear when compared with flat rolls. In an attempt to maintain overall simplicity in roll geometry, the complexity of section rolling mills has been increasing over the years, involving the application of various types of universal stands.

Conditions in groove rolling are clearly more severe when compared with the rolling of flat sections. The diversity in temperature, pressure and stress fields, along with slip gradient in calibre rolling result in accelerated wear of grooved rolls. On account of this Corbett [4] has queried whether costly secondary rolling mills (with calibre rolls) could be replaced by continuous casting, as has been the case for primary (blooming–slabbing) mills. Present manufacturing trends indicate, however, that the use of grooved rolls cannot be avoided and, indeed, various combinations of trains with calibre respective plain rolls will be applied. Even after the introduction of universal intermediate and finishing mills, there is a need for calibre rolls upstream in the rolling process. Currently there is a tendency to increase drafts at secondary mills to compensate for the absence of primary mills [4,5]. In response to these various demands, it is important to improve the resistance to wear of roll materials used in structural mills.

In general, rolls contribute some 5–15% of overall production costs, i.e. A$ 1–5 per ton of rolled product

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and per rolling mill stand. However, interruptions of the production process – shut-downs on account of wear-related problems – cause additional significant expenses [2,6]. Considering the relatively high proportion of roll costs in overall production costs, the ability to predict roll performance, especially in the domain of wear, becomes more important.

A principal aim of this present review is critically to discuss accumulated knowledge relating broadly to the question: "how does the hot wear of calibre rolls occur?" An area of specific interest concerns abrasive wear at elevated temperatures. The review is confined to hot wear of grooved rolls for producing the medium and heavy structural sections.

A further aim of this survey is to provide direction and focus to an experimental program. Roll wear factors and modes (based on their perceived significance) are reviewed and an outline provided concerning the methodology and models applied in examination of these phenomena. The research proposals and pilot experiments conducted using a two-disc rig and a scratch tester are rationalised and described.

2. Variables and modes of roll wear

2.1. Roll wear criteria and rates

The properties required of rolls differ in detail at the various stages of the rolling line. While at the initial stage of rolling process the heat shock resistance and the general strength of the material are the dominating requirements, at later stages, resistance to abrasive wear becomes the most important necessity.

In addition, there are different perceptions of "roll wear" amongst rolling mill operators, designers and roll producers. Roll "performance" is a vague term that may concern: mass or length of hot steel produced per millimetre of machined roll diameter, or per mass of roll material used; or total tons of the product processed per set of rolls or per roll change. Calibre rolls have wear resistances of about several hundreds tons of hot rolled steel per millimetre of machined roll diameter (ton mm^{-1}), whereas typical flat rolls have "lives" of about 5×10^3–20×10^3 ton mm^{-1} [1,7,8]. Such a variety of roll wear criteria makes objective comparison of roll performances difficult. This problem might be overcome by defining roll wear in terms of "radial wear (mm) per sliding path" [7].

Non-uniform wear (Fig. 1) should be observed, because the points of extreme wear determine the total depth of subsequent machining [9]. Generally, differences in wear (e.g. along the groove meridian, among the roll calibres, between the top and bottom roll, among the different stands) decrease roll life and adversely affect the rolling process and product quality.

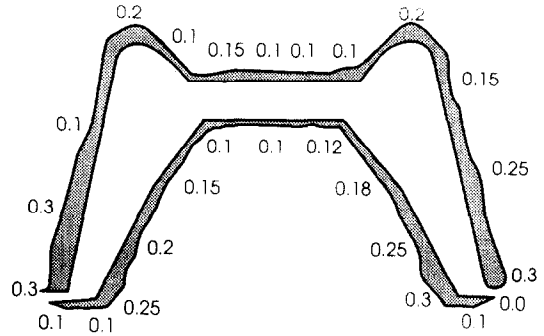


Fig. 1. Wear of the ∅650 mm rolls for 200 mm channel section [9].

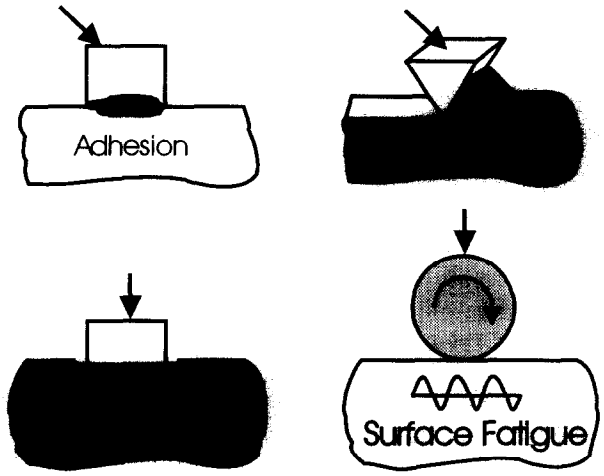


Fig. 2. Schematic diagram of the four main wear mechanisms [11].

2.2. Roll wear modes

Before arriving at a selection methodology for materials and/or other system parameters to mitigate roll wear, it is useful to identify what wear processes are possible. Various interactions can affect the wear modes, e.g. the loss or transfer of surface material may lead to loss of fit or alignment (in addition to generating debris, which is a potential abrasive), with resultant changes in loads and friction, generating subsequent wear damage [10]. Experience has shown that roll wear rate increases rapidly after production of a specific amount of rolled steel; hence roll changes should be conducted after rolling a characteristic tonnage, to avoid catastrophic wear.

To identify wear processes, it is necessary to refer to some of many published wear classification systems. An example of the useful approximation to wear categorisation is shown in Fig. 2 [11].

Abrasion is among the dominant ever-present components of the total roll wear process [12,13]. Generally, abrasion can be conceptualised as two-body or three-body wear. Considering the common presence of oxide scales of high hardness and low plasticity on hot steel surfaces, three-body wear would be expected to be an important mode of roll wear [6,11]. Further sub-clas-

sification distinguishes micro-ploughing, micro-cutting, micro-fatigue and micro-cracking [11]. Depending on the shape and hardness of the abrasive particles, there occurs either micro-cutting or plastic deformation of the surface and sub-surface layers by sliding; sub-surface deformation can result in crack formation [14]. Such a sub-classification generally has not been recognised in research of roll wear, and hence a further specific analysis of abrasive mechanisms is needed pertaining to this field.

Thermal fatigue is widely known as a mechanism of roll wear [6,12,15,16]. Any point on the roll surface is alternately heated by contacting hot steel and cooled by water. Consequently, compressive and tensile stresses are generated at a frequency of the roll rotation. If the compressive stress exceeds the compressive yield limit of the roll material during the heating stage, the outer layer will deform plastically. In cooling, a high tensile stress will be imposed on roll surface. The ductility of the roll material is significantly decreased at these lower temperatures. When the fatigue limit of the material is reached, crack nucleation begins and the characteristic overall "firecrack" pattern will result [6,16]. Heavy bending stresses will accelerate the process of crack propagation.

Various forms of *corrosion attack*, e.g. *fatigue* or *stress corrosion* are also responsible in some measure for roll wear [12]. In contrast, Ohnuki [17] has highlighted the benefits of oxides formed by high temperature oxidation at roll surfaces: at specific temperatures, dependent on the roll material grades, a hard and smooth black magnetite scale film is formed on the roll surface and excellent resistance to abrasion wear obtained.

Adherence ("*seizure*") of bands of oxide layers derived from rolled steel surfaces has been reported by some authors [17,18]. This phenomenon defined as "the transfer of part of the work material surface to the roll surface, causing loss of product quality" occurs more readily at high reduction passes. Intermittent seizure leads to hot-rolled material "pick up", roll surface roughening and increase in friction coefficient. Occasionally, associated uniform "prints" can be identified as periodic damage along the finish product surface.

Rolls can wear through slip during initial passes and hence the bite must be improved by increasing friction or by decreasing the reduction per pass. Knurling, produced on grooved roll surfaces to increase the angle of bite, can be rubbed out by skidding and the biting ability can thus be adversely affected [4,19].

2.3. The classification and properties of roll materials: selection of the roll materials

Roll materials can be classified in several ways.

2.3.1. Roll manufacturing route

The roll manufacturing route may be via casting, forging, double pouring, etc. Spin casting is used today by the majority of manufacturers of rolls to be used for flat products. In contrast, rolls for heavy structural products require grooves so great that a useable layer thickness could only be obtained by a roll involving a single pour. With the development of sleeves for universal stands, spin casting has also been applied for structural mills and, today, most rolls for universal stands have been centrifugally cast [20]. Universal mill finisher roll sleeves (channels and beams) are press-fit, glued or shrunk onto the shaft. Stresses to be induced by the fit will be influenced by the roll material choice. Spheroidal irons and high carbon steels are generally adequate for both horizontal and vertical rolls. Some forged and composite vertical sleeves are in use [4]. Yanaka [4] describes the utilisation of sectioned sleeves: with these composite sleeved rolls it is possible to simultaneously improve both the wear resistance of the working surface (sleeve is made from a high-hardness material) and the breakage resistance of the arbor (high-toughness material).

2.3.2. Roll material chemistry

The range of chemical analysis is: $0.3\% \leqslant C \leqslant 3.8\%$; $0.2\% \leqslant Si \leqslant 2.5\%$; $Mn \leqslant 2.5\%$; $Cr \leqslant 30\%$; $Ni \leqslant 5\%$; $Mo \leqslant 4\%$.

Further alloying elements are P, S, V, W, Nb, Ti. The contents of P and S are carefully controlled [4,19,21].

2.3.3. Microstructure of the roll working surface

Depending on chemical composition and heat treatment, roll materials have a microstructure consisting of a matrix of pearlite/bainite/martensite/austenite, with a greater or lesser proportion of carbide/cementite, and graphite of a particular distribution and size [19]. The properties of roll materials are significantly influenced by the quantity and distribution of carbides (M_xC_y) which can comprise up to 40 vol.% of the roll microstructure. Cementite has exceptional abrasion resistance at room temperature and excellent wear resistance at elevated temperatures. In general, there is a tendency to strive for the maximum quantity of carbide in the roll structure compatible with adequate ductility, strength and heat shock resistance [4].

Roll materials with graphite (flake or nodule) have generally improved tolerance of indifferent cooling conditions (i.e. enhanced firecrack resistance) and reduced friction in designs where thrust collars and/or side wear are critical.

Classical categorisation has divided roll materials into two main groups: (i) cast iron rolls and (ii) steel rolls. Their essential differences can be summarised as follows.

(1) Cast iron rolls (often used at finishing stands) are brittle materials that possess high wear resistance.

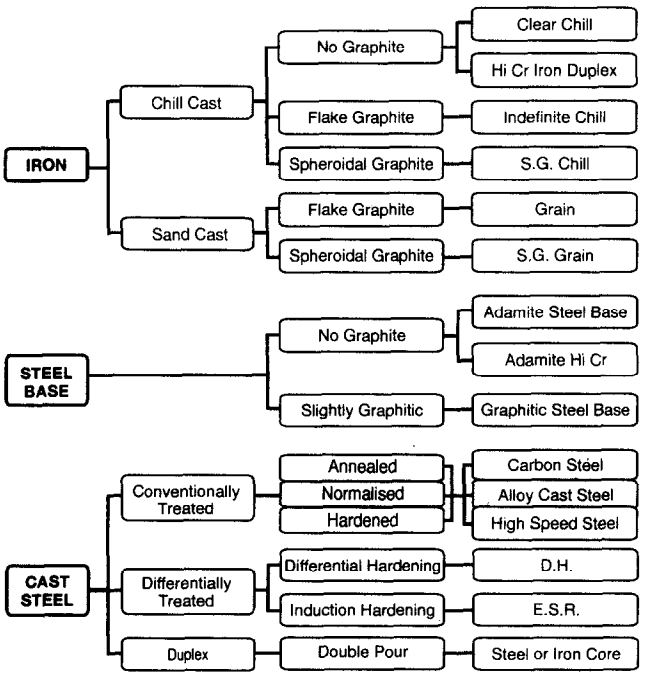
(2) Steel rolls are often used in primary and secondary mills because they exhibit the following (superior) properties in comparison with cast iron rolls: "higher" coefficient of friction (leading to a better "bite"); "higher" strength, so they can withstand "higher" bending and torsional stresses; and uniform hardness to a sufficient depth from the surface [4].

Recently, highly alloyed ferrous materials, notably the high chromium irons and steels, as well as high speed tool steels, have been promoted as roll materials with extremely good wear resistance [5,22–24]. Table 1 shows a classification of roll materials based on the structure of the working layers.

Categorisation of roll materials can be based on an understanding of the interrelations of mechanical properties, microstructures and chemical compositions, especially C content, see Fig. 3 [5]. The most important metallurgical parameters that influence abrasive wear of steel are microstructure (i.e. type of matrix and dispersed phases), hardness at working temperature and interstitial element content (%C and/or %N) [25].

When considering the selection of roll materials, in relation to general wear problems, more emphasis is often put on improving the strength and hardness of metallic alloys, rather than their ductility [11]. However, ductility can decrease significantly with increasing strength or hardness. Roll cracking susceptibility due to thermal fatigue and abnormal thermal shocks are mainly affected by fracture toughness, thermal conductivity, thermal expansion and ductility. Hence, a compromise between strength and ductility has to be looked for.

TABLE 1



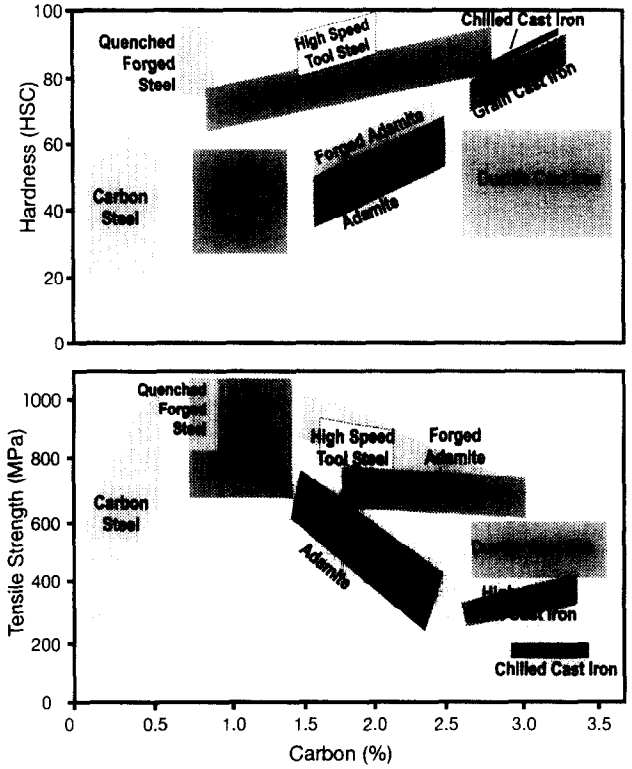


Fig. 3. Tensile strength and hardness of materials.

2.3.4. Selection of the roll materials

The practical importance of the tool material can be outlined if specific problems in roll choice are considered. For example, to select materials for rolls in a universal medium section mill, the following criteria were applied: 1. The roll surface defect history (thermal cracks or other wear modes); 2. Product categories (light, heavy); and 3. Mechanical properties required (stresses expected). Generally, materials offering the best combination of resistance to mechanical stresses, thermal fatigue and wear, are various centrifugally cast steel base grades (adamites). Thus adamites were chosen commencing with a material with a hardness of 300 HB for the entry group of stands, and ending with 420 HB for the finishing rolls. Further improvements were sought by use of high Cr alloy cast irons and alloyed bimetal centrifugally cast spheroidal irons [26].

A quite different situation appears in mills producing heavy structural products, where the loads are high and the roll geometry is such that stresses become severe as the roll diameter is reduced. For the breakdown rolls, toughness, as well as resistance to thermal shock and wear, is required [27]. Here it is apparent that there is a conflict in material requirement and, generally, mono bloc steel-based rolls are selected. In these situations material commonly ranges from cast

carbon steel (e.g. 0.5% C, with hardnesses of 30 Shore C, tensile strength 650 MPa and ferrite–pearlite structure) to adamite rolls (e.g. 2.3%C, Cr–Mo alloyed, with hardnesses of 50 Shore C, tensile strength 500 MPa, and a microstructure of spheroidised carbides dispersed in a pearlitic matrix) [4].

In some modern mills, spheroidal iron is favoured for the intermediate stands, for it exhibits the strength of the higher carbon steel roll grades and the wear resistance of the alloy iron grades. For the same intermediate stands, but with deep groove rolls such as angles and channel rolls, a high carbon steel roll or even a graphitic steel would be a better choice. The reason for this is that a steel roll can be heat treated to maintain a uniform hardness to a sufficient depth – much greater than it is possible with either the spheroidal iron or alloy grain irons. Grain iron roll material ("indefinite chill") has an outer chilled face on the roll body with a significant fraction of finely divided graphite flakes solidified in the surface layers. These graphite flakes increase in amount and in size as distance from the surface increases [28]. The structure of alloy grain irons depends critically on cooling rate during casting and they generally lose hardness very rapidly from the surface to a depth of about 80–100 mm. Hence, a roll made from iron will wear non-uniformly [4].

Rail and large structural intermediate stand rolls are generally always fabricated from steel, although some success has been achieved using pearlitic acicular spheroidal iron rolls [4].

2.4. Rolling loads and stresses

Hot wear rate is directly proportional to the normal pressure on roll surface. Average rolling pressures can be considered to be in the range 100–300 MPa. The corresponding cyclic stresses, amplified by thermal cycles, in roll surfaces are estimated to amount to ± 500 MPa [7,17].

Cyclic loads result in material fatigue and other forms of surface deterioration. Besides the well-known mechanisms of cyclic softening and corrosion fatigue, there is growing evidence of the damaging influence of tensile stress during the contact fatigue, leading to cracking and pitting [29]. In addition, it has been found that cyclic pre-stressing has a significant influence on the material removal process in sliding wear [30]. A related question is the actual distribution patterns of the rolling loads and stresses. The observed non-uniform wear, particularly the unsymmetrical wear of the symmetrical calibres, illustrates the significance of the irregular pressure distribution. The appreciation of these contrasts in stresses has led to the introduction of different materials for top rolls and bottom rolls in the practice of section rolling.

If metal flow is not appropriately allowed for in roll pass design, metal is unnecessarily forced to exert additional localised pressure and wear on the groove walls. There are in fact basic principles for roll pass design in existence, where roll wear is among the main criteria. Higher loads and draft non-uniformity should be applied in the initial passes and they ideally should diminish in the finishing passes [2,31]. Rolling loads and stress distributions should be designed to ensure stable and uniform wear of finishing calibres. Corrosion fatigue can be suppressed by proper design considerations aimed at reducing stress. The tensile component of stress causes stress corrosion. Residual compressive stresses, deliberately introduced through an adequate heat treatment, will suppress crack initiation and growth, as well as stress corrosion and fatigue [4]. Geometrical aspects (e.g. roll diameter and small meridian radii) have considerable effect on tool life via associated stress and heat concentration [7,32].

Considering the feasibility of influencing stresses and loads as process variables that can be controlled by roll pass design, this established correlation between stresses and surface deterioration must be an important aspect of research into roll wear.

2.5. Aspects of rolling temperature

Rolling temperatures vary mainly between 800 and 1200 °C [7,17,18]. The roll surface is heated initially to approximately 650 °C while it is in contact with the hot slab, and subsequently cooled by water to around 50 °C during the same cycle [27]. The flash temperature could in fact rise above 800 °C owing to the friction-generated heat [33].

An important aspect is the effect of surface temperature on the formation of oxide scales on roll surfaces. Cast steel and adamite develop hard and smooth magnetite black scale films above 400 °C; this process is followed by a rapid decrease in wear and the friction coefficient f. As the surface temperature rises, the scale transforms from magnetite to wustite. Above 650 °C and higher, the surface layer matrix is softened and deformed plastically. Hence, cast steel and adamite materials should be used in the temperature range 500–650 °C. For analogous reasons, grain cast iron rolls should be used between 600 and 700 °C. This is because the eutectic carbide and primary crystallised graphite in the surface act to halve the effective area available for magnetite scale formation. Above 600 °C, the wustite scale from the counter-body adheres to the cast iron to form a black scale, thereby reducing wear rate. At 750 °C and above plastic deformation occurs in the surface layers, increasing the surface roughness [17].

An additional phenomenon associated with temperature is thermal fatigue [16,17,27]. At higher temperatures in the use of roll materials (e.g. 600 °C), acceptable

stress levels in fatigue are generally lower when compared with behaviour at room temperature. By the use of a counter-rotating discs test, Ohnuki [17] has examined the surface-layer thermal fatigue of hypereutectoid cast steel and adamite. At contact stresses of about 250 MPa and roll surface temperature cycles of 100–500 °C, the thermal fatigue limit was found to be in about $(1–4)\times10^4$ rev. The surface layer fracture originated at a depth 0.1 mm below the surface; that is where the contact shear stress reaches a maximum, and this effect terminates the protective action of the oxide scale.

Roll surface temperature depends also on the deformation zone geometry. This again indicates the importance of roll pass design.

2.6. Cooling and lubrication

Although it is clearly understood that temperature variation around and across the roll surface is detrimental, extremely wide differences in cooling practice remain unsatisfactorily explained [16]. Water is widely used as a coolant in section rolling; it is applied either on its own or within an aqueous dispersion or emulsion [1]. Experience in rolling practice has demonstrated that idle rolls should not be cooled, because of the damaging effect of the increased thermal gradient. A more sophisticated problem is the optimisation of cooling the calibre rolls via modelling, hydrodynamic, thermomechanical and tribological parameters.

Lubrication is widely considered as a method for the improvement of roll life. Mineral oils as well as organic fats and oils were tried as lubricants in the early stages of rolling technology development. Light mineral oils (compounded with additives such as lanolin) exhibited low staining propensities. In the hot rolling of steel plates, heavy and possibly contaminated mineral oils were initially used: this practice was understandably abandoned [1]. Many authors confirm that significant decreases in roll wear have been achieved by use of various special lubricants, although some limitations are reported relating to their use at high temperatures (above 900 °C) and the influence of hydrocarbon fluids, in particular, on roll performance. Several authors, however, indicated that, in hot rolling of steel, lubrication sometimes paradoxically increases roll wear [1,34,35]. Schey [1] has discussed some aspects of the health and environmental risks, associated with lubrication practice in hot rolling.

Williams [12] has discussed the possibility of roll cooling by air and indicated that glass might have attractive lubricating properties.

2.7. Hot-rolled steel

The chemistry, mechanical properties and dimensions of the rolled product undoubtedly all have a strong influence on wear. Particularly rapid wear of calibres occurs during the rolling of alloy steels. This is partly explained by an increase of deformation resistance in the work piece, especially during rolling at lower temperatures [9].

Magnee [7] and Sheasby [36] have discussed details of the environment of the deformation zone especially concerning oxide layers present and their effect at rolling temperatures, thus: (i) haematite (Fe_2O_3) of hardness 1050 HV subjects the roll surface to a severe abrasion; (ii) magnetite (Fe_3O_4) generates abrasive particles of hardness 450 HV; (iii) wustite (FeO), of hardness 300 HV, acts as a lubricant.

The observed effects of temperature on wear of roll materials via oxide influence may be explained as follows: (i) the oxides formed in the range 400–600 °C are Fe_2O_3 and Fe_3O_4 and increase abrasive wear; (ii) in contrast, within the range 600–900 °C, the progressive formation of wustite increases the lubricating effect of the scale and effectively decreases roll wear; (iii) at temperatures above 900 °C, the formation of magnetite and haematite leads again to an increase in abrasive wear [7].

2.8. Relative slip and sliding distances

Increase of wear loss with relative slip and sliding distance is a proven effect. Detailed studies of metal flow during rolling of rectangular profiles have revealed the elaborated laws of metal flow in this, relatively simple, case of rolling [1,2,31]. In calibre rolling, the situation is more complex owing to variations of all wear factors along the groove meridian and the resulting non-uniform wear. For example, the differences of roll diameter in the axial direction, amplified by variances in area reduction, cause a significant gradient of relative slip along the groove width. Wear in calibres is increased further by metal flow, perpendicular to the rolling direction [31]. Various pass sequences have been developed in attempts to minimise these variations and optimise the roll wear resistance. Fig. 4 depicts the variety of roll designs used in rolling of channel sections [9].

Last, but not least, it should be noted that the rolled bar is much longer at finishing stage when compared to the break-down stage of rolling. In roll pass design practice, this is compensated by a corresponding distribution of loads and stresses.

3. Methods and models used in roll wear studies

3.1. Aspects of wear testing

The "holy grail" of engineers is a rapid, inexpensive, accelerated laboratory wear test, which will accurately

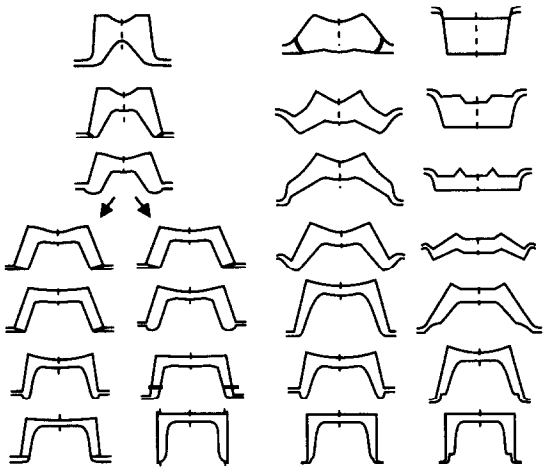


Fig. 4. Roll pass designs for channel sections [9].

predict the performance of materials for very specific applications without expensive industrial trials. On the other hand, scientists are looking for a test method that entirely isolates a given wear mechanism, or variable, from all others [37].

The wear literature contains literally hundreds of different machine designs, and it would appear that the number of "one-of-a-kind" purpose-built machines far exceeds the number of "standard" machines. Various systematisations are attempted, e.g. DIN 50320 classifies wear tests by the type of "tribological action", which leads to a list of 13 tribo-systems involving sliding, rolling, impact, fretting and abrasion in various media and motions [37].

Blau [37] suggests the following questions when a wear testing device is to be designed: 1. To what use will the test data be put (basic research, quality control, simulation, materials selection, standard development)? 2. Do the needed data already exist (is it cheaper to purchase these data or to carry out one's own testing)?

Several parameters should be used to measure wear during the same experiment. Generally, wear tests exhibit significant variations in reproducibility [37].

Other methods relevant to background data collation are: chemical analyses, mechanical testing at room and/or elevated temperatures, thermophysical property tests, metallographic examination, corrosion tests and non destructive tests.

A *roll wear test* is frequently conducted by measuring the loss of sample weight per unit of surface area after the sample has been in moving contact with a standard hardened surface under defined conditions (temperature, pressure, time, speed). The contact might be sliding or rolling with or without abrasive, and one material can be tested in contact with identical, similar or dissimilar materials. Although such laboratory wear tests attempt to, but rarely mirror service conditions, they have been used for ranking materials. Thus, while different types of wear test may permit classification of roll materials, there is always the need to compare wear test results with performance from the real service life of actual rolls [4].

A significant number of authors have used a two-disc wear test to simulate roll wear at high temperatures (Fig. 5). A two-disc apparatus enables a separation of three essential aspects in roll wear investigation: load, slip and temperature. This device enables simulation of the following ranges of the key variables: specific working pressure, 50–250 MPa; sliding speed, 0–3 m s^{-1}; temperature of "hot steel" disc, 25–900 °C; thermal cycling options exhibit a satisfactory flexibility [7].

One of advantages of this two-disc method is the ability to simulate cyclic wear involving long sliding distances. The degree of similarity with hot rolling is in fact sufficient for many purposes (e.g. evaluating roll materials, assessing the role of lubricants, etc.). Obviously the costs and running times of such experiments are much lower than they would be for even semi-industrial trials. However, the method is not without problems. The main disadvantages are: 1. the stress state in the roll material is only a rough approximation to real conditions; 2. the ratio of elastic to plastic deformation of the counter body is much higher than in the real situation; 3. with increase of plastic deformation there is a corresponding decrease in the ease of cyclic testing over long sliding distances; 4. Lundberg [6] has stated that the test roller is exposed to heat radiation from the heated disc during the heating/idling period, unlike in the real rolling process; this is, however, because of Lundberg's particular design of hot wear rig: the heating of the "steel" disc ∅500 mm×45 mm by the propane burner takes approximately 30 min.

Matsuda et al. [22] and Goto and Mase [38] have conducted a block-on-ring test in addition to the two-disc method. The counter-body materials (stainless steel and carbon steel) were heated to 800 °C by the induction coil. The wear of graphitic cast iron was examined and the following conclusions were reached: (i) the coefficient of friction generally decreased when the amount of graphite in a structure increased; (ii) the wear initially

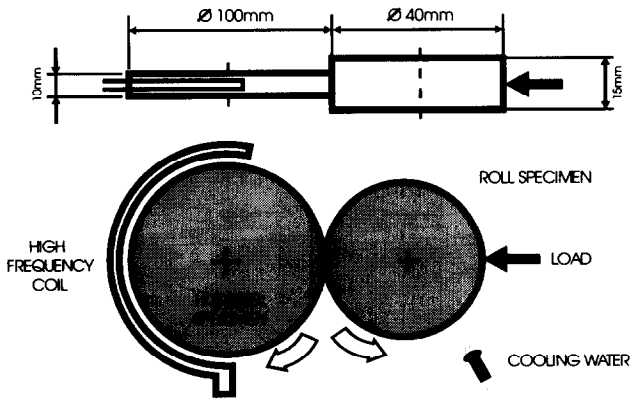


Fig. 5. Schematic illustration of disc-on-disc wear test.

decreased but subsequently increased with the amount of graphite in the microstructure, except when the counterface material was carbon steel (S45C) under low stress; (iii) for carbon steel and low stress conditions, wear increased regularly with the amount of graphite.

Some authors have examined the wear resistance of roll materials by using a small, laboratory scale, hot rolling mill (e.g. one-tenth size of a production mill roll diameter). Such an experimental mill gives the closest approximation to actual industrial conditions. These tests are of course significantly more expensive to conduct than are other tests [22,24,34]. Matsuda et al. [22] have effectively adopted these three wear tests to rank roll materials: two-disks, block-on-disc and laboratory hot rolling mill.

Among the scientific methods focused on the assessment of abrasive wear, the scratch test is an example of an attempt to isolate one wear mechanism from others [39]. Such scratch tests allow studies on abrasion without the super-imposition of other major wear mechanisms such as surface fatigue or adhesion. This method has been used for the investigation of wear behaviour of hard multiphase metallic alloys subjected to abrasive wear at elevated temperatures up to 1000 °C [39].

From the viewpoint of hot roll wear, the two-disc testing device and the scratch tester can be considered as two extremes in the experimental approach. The former is a comprehensive method focused on simulating the real situation as closely as possible. The latter technique, single-point scratch tests, enables isolation and identification of specific wear mechanisms and particular wear factors.

3.2. Wear models

Lim and Ashby [40] state that for each of the major wear mechanisms wear rate can be expressed in terms of:

$$w_i = f_i(F, v, T_0, M) \quad (1)$$

where w_i is the wear rate ($m^3\ m^{-1}$), F the normal force (N), v the sliding velocity ($m\ s^{-1}$), T_0 the initial temperature (°C) and M material properties (e.g. yield strength, MPa, etc).

Each factor from the Eq. (1) can be expressed approximately as a vector with n components:

$$v_i = (v_{i1}, \ldots, v_{in}) \quad (2)$$

and each component can be considered as a variable.

There is an obvious necessity to reduce the number of variables in Eqs. (1) and (2). Theoretical disciplines contain relations involving some of the above variables. Some of the components can be considered as discrete variables, while others can be assumed as approximately constant values. Fig. 6 summarises the factors affecting roll wear.

According to published knowledge, this problem has been approached by whole range of wear models from

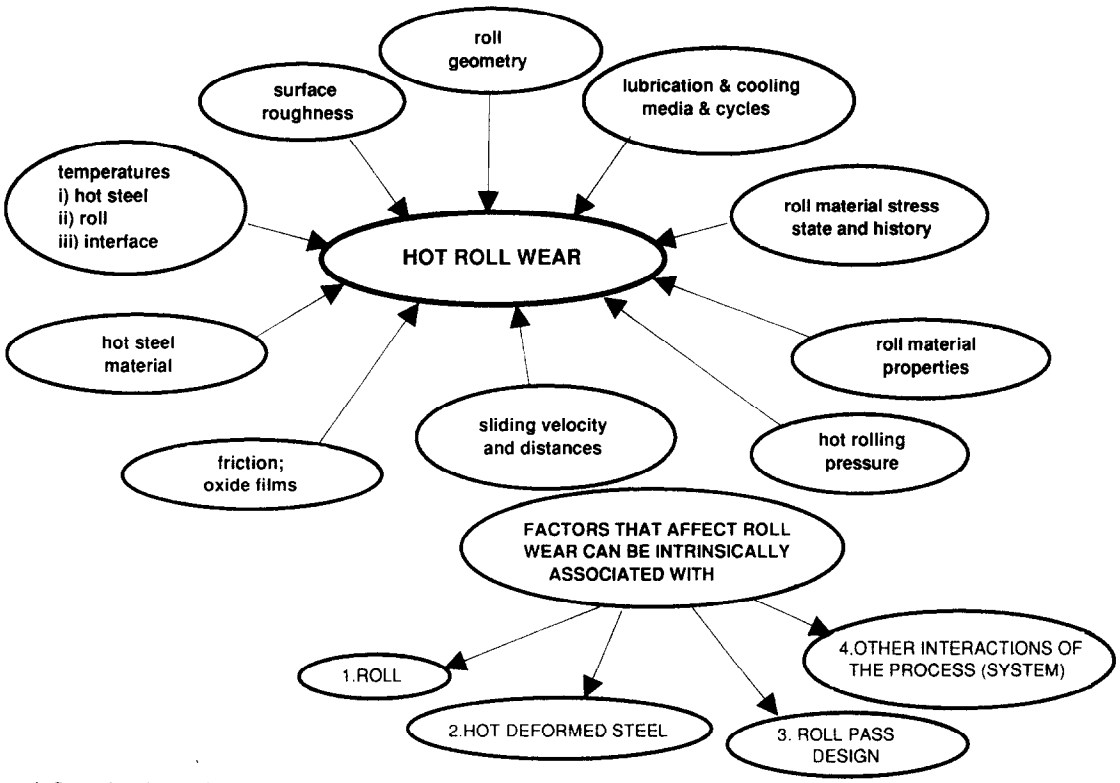


Fig. 6. Factors influencing hot roll wear.

simplified, general wear equations to "comprehensive" formulae for special cases of hot wear of roll materials. Here the criteria for optimising the number of arguments in any model can be found in mathematical statistics.

Archard-Holm's equation for worn volume during sliding–adhesive wear is an example of a fundamental approach [25]:

$$W=KFL(R_p)^{-1} \quad (3)$$

where W is the worn volume (mm^3), K the wear coefficient, R_p the proof stress (MPa), F the normal force (N) and L the sliding distance (mm).

In the open literature can be found more comprehensive attempts to model special cases of hot roll wear. Thus Felder [41] has suggested that the wear mechanism (abrasion or thermal fatigue) in rolling depends on a relationship involving the shear factor m (defined in [1]) and the deformation zone geometry.

Magnee [7] has proposed the following model:

$$\mu=AD^2(v)[(W/W_c)-1] \quad (4)$$

where μ is the wear (kg m^{-2}), A an intrinsic constant of the roll material, $D^2(v)$ a quadratic function of speed, associated with the kinetic energy of the process, W the strain energy in a roll material induced by displacement of the friction force (J), W_c the critical energy (J) (when $W>W_c$, then a wear phenomenon occurs), $W/W_c=f_1(M, F, T_0)$, $D^2(v)=f_2(V_o, V_e)$, and V_o, V_e are the hot steel velocity and the roll peripheral velocity (m s^{-1}), respectively.

Variations of the wear rates in groove rolling can be calculated by introducing in Magnee's model the relevant values of V_o, V_e, T_0 and F corresponding to their variations along the groove meridian. Magnee has demonstrated how parameter A from Eq. (4) can be used to evaluate roll materials, especially in the case of Cr–Mo addition level. A propitious change of roll material, characterised by significant decrease of parameter A, will result in the decrease of wear.

In model (4) the bulk stress state and stress history of the roll material have not been taken into account.

4. A research plan and pilot tests: rationale and detailing

4.1. Research proposals

Critical consideration of the foregoing review suggests that a useful main focus of research would concern hot abrasive wear of grooved rolls, although interactions with other wear modes should also be evaluated within a structured programme. The following aspects are worthy and appropriate to be addressed: (i) building a logical structured concept (e.g. wear rate criteria, definitions, dependent and independent variables, limitations); (ii) categorisation of roll wear factors, permitting the focusing of the research onto most significant and feasible parameters; (iii) analysis and selection of the methods to be used for simulated roll wear studies; (iv) a laboratory experimental program assessing the effect of materials, load, roll stress, slip ratio and temperature; (v) the upgrade of Magnee's model [7] for roll wear above 1000 °C and the relative slip below 0.3 m s^{-1} via statistical analyses of the data gained during step (iv); (vi) the correlation between roll pass design and roll wear.

A first hypothesis in this research program is that Magnee's model, defined by the Eq. (4), describing hot roll wear up to 900 °C, can be extended to temperatures above 1000 °C and to the relative slip below 0.3 m s^{-1} without the appearance of cuspidal points. Testing of this hypothesis will be realised by means of hot wear rig via statistical analysis of the effects and interactions of temperature, normal load and relative slip. The stochastic approach will improve the approximation to real hot wear situation when compared with the original deterministic form of the Magnee's model.

Following the analogy with stress corrosion and cyclic softening of steel materials, a second hypothesis to be tested in the program is that the bulk stress state and stress history have significant influences on abrasive wear. The effect of bulk stress on abrasive wear can be evaluated by means of a single point scratch tester. Here exists the potential to examine additional aspects of the interaction with oxide films.

A third hypothesis is that the various adopted pass sequences of rolling, published in the literature, in fact are associated with optimised roll wear resistance, because these pass sequences were developed during centuries of trial-and-error industrial rolling practice and corrected many times towards the same aim of minimised roll wear. In this context it is believed that there is a correlation between the parameters of roll pass design and the non-uniform wear of the grooved rolls.

Within this overall rationale it has been decided to use two test methods: (i) two-disk hot wear rig and (ii) scratch tester. Two types of roll materials are to be assessed: (1) a material already used in rolling practice with known wear resistance in industrial application, namely adamite; (2) high alloy steel, which may be expected to have better wear resistance than adamite.

4.2. Preliminary experiments

Some preliminary experiments were carried out using a *scratch tester*. Hypo- and hyper-eutectoid steel specimens were bent to obtain tensile or compressive stress and then scratched in situ under a normal load of 25–40 N. In harmony with the phenomenon of increased

plasticity of a material exposed to a tri-axial compressive state, the scratches produced on surfaces under compression exhibited a higher level of plastic deformation, when compared with scratches made on surfaces exposed to tension. The scratches on the surfaces under tension were deeper when compared with scratches made on surfaces under compression or unstressed surfaces. A tensile state favours the micro-cutting abrasive sub-mechanism which can account for the observed increase in scratch depth of more than 15%. The domination of micro-ploughing at surfaces under compression favours strain hardening of the worn layers. Further research is proposed on the effects of bulk temperature of the worn material [42].

Testing of adamite (1.8% C) by means of a *two-disc tester* (steel disc was K1045) has been commenced involving variations of the normal force (60–160 N) and relative slip (0.2–0.6 m s^{-1}) at temperatures between 760 and 960 °C. The sequential experiments are conducted to evaluate the resolution capability for two levels of force: 60 and 100 N at 860 °C and slip 0.5 m s^{-1}. With ten runs the testing method has enabled the measurement of the wear gradient with 80% confidence. This confidence limit can be increased by increasing the number of runs or test duration. A further series of experiments was conducted at two temperatures: 860 and 1000 °C with constant force of 100 N and slip of 1 m s^{-1}. After 20 runs it was concluded with 90% confidence that wear did not decrease with an increase in temperature from 860 to 1000 °C. Other authors [7,41] have claimed that wear does decrease with increases in temperature between 700 and 900 °C.

A separate group of pilot runs, made with clockwise rotations of both discs, was compared with a situation in which the roll disc and the steel disc were rotated in opposite directions. The treatment using clockwise rotation of both discs exhibited a significantly higher wear rate. This phenomenon would explain the increased wear in some calibres during the section rolling, where the particular components of metal flow have the direction opposite to the corresponding components of the roll velocity. More testings are required to confirm this phenomenon.

Further tests are programmed by means of factorial design, to evaluate the significance and interactions of main wear parameters.

5. Conclusions

Steel rolling is recognised as one of the most important industrial processes. Rolling using grooved rolls (as a category different from grooveless rolling) is the most common practice in production of steel sections. Key tools in this process are the rolls that contribute up to 15% of production costs. A main cause of roll consumption is continuous wear, a complex process where mechanical and thermal fatigue combines with abrasion and corrosion. The necessity to compensate for the non-uniform wear during machining is an additional aspect of roll consumption. An area of specific interest is concerned with abrasive wear within the environment of rolling in grooves, where the nature of the deformation zone can accelerate roll surface deterioration.

The published research into roll wear is mainly concerned with the effects of roll material, oxide scales, temperature, normal force and sliding velocity. Though a selective application of numerous roll materials is decisive for roll consumption, there is no general index for determining their resistance to wear. Several authors have examined how temperature (up to 1000 °C) has affected the wear of various roll materials via corrosion, formation of oxide films and fatigue. Detrimental influences of the rolling load and relative slip on hot wear are clearly proven. On the other hand, the published works give very little, if any, information on the possible ranking and interactions of wear factors. In addition, there is a lack of knowledge about the effects of hot steel temperatures above 1000 °C and the influence of bulk stress in roll material on abrasion.

The published wear models indicate that a deterministic approach dominates in applied mathematical definitions. Wear equations are based on deterministic relations of heat transfer, diffusion, plastic deformation and tribology. These published models of roll wear are mainly focused on the relatively simple case of flat rolling and there is no information relating to possible statistical interactions and variations in wear. A literature search has revealed no stochastic approach in modelling wear.

Roll pass design exerts a vital influence on the non-uniform wear of rolls via control of loads, stresses and relative slip. In addition, the groove wall inclination has a direct influence on the radial machining required to regenerate the groove geometry. The metal flow in grooves is determined by the interactions of subsequent stages of the whole process. An important question is: why does a particular location on the roll experience more wear than others, when the whole surface of the roll is made from the same material?

The pilot experiments carried out by means of a scratch tester have proven that abrasive wear differs depending on whether the specimens are under tension or compression. Steel material exposed to bulk tensile stress shows a decreased resistance to abrasive wear.

In experimental testing of roll wear, the two-disc devices are widely used in various configurations. There are certain advantages such as extended test duration and stability within the Hertzian stresses. However, it has been reported that there are some limitations: the

method simulates hot wear in flat rolling only and there are certain difficulties regarding the use at temperatures above 1000 °C. Pilot trials have confirmed the device reliability, especially for ranking of materials by exposing them to identical conditions of hot wear simulation. With a recent improvement of the method, the experiments above 1000 °C appear to be promising.

The use of other techniques is still widely reported. To improve the reliability of ranking roll materials, at least two separate testing methods are to be applied, along with the appropriate standard metallurgical examinations.

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